Ajay Kumar Patro

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GIS Backend & Geospatial Data Engineer — PostGIS, Python, GeoServer, Keycloak

Technical Skills

Languages: Python, SQL, JavaScript, Bash
Frameworks: Django, FastAPI
Databases: PostgreSQL/PostGIS, MySQL, SQLite
Database Engineering & ETL: Spatial SQL, Query Optimization, Spatial ETL, Indexing, Schema Design
GIS:
Tools: ArcGIS Pro, ArcMap, QGIS, PyQGIS, ArcPy, GDAL/OGR, Orfeo ToolBox, Geopandas, Google Earth Engine, PDAL
Services: GeoServer, Martin, pg_tileserv, TiTiler
Infrastructure & Tools: Docker, Docker Compose, Linux, IIS, Apache, Tomcat, Nginx, Git, GitHub
Backup & Auth: pgBackRest (PITR), Cloudflare R2, Keycloak/OIDC

Experience

GIS Analyst
Mar 2026 – Present
Monarch Surveyors & Engineering Consultants Ltd — *Nashik, Maharashtra, India*

- Deployed a 2-node **GeoServer** cluster for a state government agency behind **IIS** (reverse proxy) with **Apache Tomcat**, **Keycloak**/OIDC and HTTPS/SSL, using an **SMB** share over the existing **SAN** as a common data directory.
- Wrote a **batch script** to deploy primary and secondary nodes, remove existing installations, and promote either node to primary, enabling manual failover in 10–20 minutes, within the client's 1-hour downtime tolerance.
- Designed and set up PostgreSQL/PostGIS backup and recovery from scratch using **pgBackRest** with **Cloudflare R2**, with weekly full backups and **point-in-time recovery (PITR)** for a ∼9 GB database; adopted by the team.
- Managed **PostgreSQL/PostGIS** and **GeoServer** for a multi-tenant LRMS platform supporting government land acquisition and infrastructure projects, including spatial schema design and ingestion, transformation and validation of survey, drawing and `.shp` datasets.
- Converted drone imagery from `.ecw` to Cloud Optimized GeoTIFFs (COGs) with internal overviews and quantized alpha bands, and published raster and vector layers through **GeoServer** (WMS/WFS/ImageMosaic) for dashboard web maps.
- Fixed geometry, SRID and topology problems in **PostGIS** and debugged spatial SQL, raster rendering and data-sync issues across **PostGIS** and **GeoServer**.

GIS Specialist (Python Data Ops Automation)
Apr 2024 – May 2025
Lepton Software Export & Research Pvt. Ltd. — *Pune, Maharashtra, India*

- **Project:** GOFNMS Tele Infra (telecom network data processing)
- Populated and maintained fibre-network records in the project database from stakeholder data: fibre direction, inside-plant (ISP) data, and subscriptions mapping each fibre to its **OTU/IPS** port.
- Reconciled live **OTU/IPS** fibre data against the database, reviewed fibre-length mismatches flagged by **Enghouse NetDesigner** (ArcGIS-based), and reported them to the team responsible for corrections, supporting fault analysis.
- Created inside-plant (ISP) records from field network engineers' reports to support telecom network inventory and planning.
- Built Python (**ArcPy**) automation for daily spatial data processing, validation, and reporting.

Scientific Intern (C-GARD, NIRDPR)
Mar 2023 – Jul 2023
NIRD & PR NERC — *Guwahati, Assam*

- Maintained and validated geospatial datasets for infrastructure development planning, including ground assessments aligned with **RADFI 2017** guidelines.
- Delivered technical tutorials on the **Yuktadhara–Bhuvan Portal** to field officers across Assam and Meghalaya.
- Acted as a secondary-level reviewer for research papers in the ISBN-published C-GARD book, ***Geoinformatics Application in Land, Water, and Infrastructure Development***.

Projects

Secure Vector-Tile Serving Platform: Martin + APISIX + Keycloak (In Progress)

- **Tech Stack:** *Martin, PostgreSQL/PostGIS, Docker Compose, Apache APISIX, Keycloak/OIDC*
- Building a containerized service that serves vector tiles from **PostGIS** through **Martin**, with **Apache APISIX** enforcing **Keycloak**-based role and resource authorization before requests reach the tile server.
- Designing cached authorization decisions to keep per-request validation fast, and working out how cached tiles and decisions should stay correct when data or roles change.

End-to-End AI Burn Scar Detection Pipeline (Ongoing)

- **Tech Stack:** *PyTorch, Apache Airflow, STAC Catalog, FastAPI, PostGIS, GeoServer, Rasterio, GeoPandas*
- Developed an automated pipeline to ingest Sentinel-2 imagery via CDSE's **STAC API** to detect wildfire burn scars using the HuggingFace Model: ibm-nasa-geospatial Prithvi EO-2.0 Model.
- Orchestrated end-to-end **ETL** workflows with Apache Airflow for preprocessing, inference, and spatial data loading into **PostGIS**.
- Generated STAC-compliant metadata and published GIS-ready spatial outputs via **GeoServer**, exposing them through **FastAPI** as REST API endpoints for downstream analytics and visualization.

Spectral–Index Feature Engineering / OBIA Pipeline

- Built an end-to-end **Object-Based Image Analysis (OBIA)** pipeline for satellite imagery, reducing land-cover mapping time by ∼**92%** by deriving spectral-index–based object features for automated training label generation. Implemented using **QGIS Model Builder** and **Orfeo ToolBox**.

LiDAR Point-Cloud Web Viewer (Potree, Range-Aware HTTP Streaming) Source Code

- Built a web-based LiDAR point-cloud viewer using Potree for large-scale multi-million point visualization. Engineered a range-aware static HTTP streaming server to replace Django static serving, enabling level-of-detail (LOD) octree loading, eliminating repeated full-file downloads, and optimizing browser memory and rendering performance. Implemented using Django, Potree and Flask.

Football Field Booking Management System (CS50 SQL Capstone) Source Code

- Designed a normalized, ACID-compliant **MySQL** schema (10+ tables) with constraints, triggers and indexing; tuned queries with EXPLAIN.

Education

Central University of Karnataka Nov 2021 – Aug 2023
M.Sc. in Applied Geography & Geoinformatics *Kalaburagi, India*

Coursework: GeoAI, **LULC** classification (supervised and unsupervised), **hydrological** and basin morphometric analysis (hypsometric integral, **HRI/HRS**), soil erosion modeling (**USLE**, **MUSLE**), **geostatistics** (hotspot analysis, spatial autocorrelation), geocoding, reverse geocoding, and **multi-criteria decision analysis** (MCDA) for site suitability.

Certifications

CS50 Python Certificate View Certificate
Harvard's OCW CS50, Cambridge, Massachusetts 2023 – 2024

CS50 SQL Certificate View Certificate
Harvard's OCW CS50, Cambridge, Massachusetts 2025

ArcGIS Enterprise MOOC
Esri 2026 – Present